NAME _____ CLASS _____

BWAILA CLUSTER MOCK EXAMINATIONS

2022 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

BIOLOGY

Subject Number: M022/I

Tuesday, 24th May, 2022

Time Allowed: 2 hours
8:00 – 10:00 am

PAPER I

(100 marks)

Instructions

1. **This paper contains 11 printed pages. Please check.**
2. Fill in your **name** at the top of each page.
3. This paper contains **three** sections **A, B** and **C**. Answer **all** the questions in the spaces provided.
4. Use of electronic calculators is allowed.
5. The maximum number of marks for each answer is indicated against each question.
6. In the table provided on this page, **tick** against the number of the question you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			

Turn over

SECTION A (70 marks)

Answer all questions in this section.

1. **Figure 1** is a diagram representing gas exchange between a leaf and the atmosphere. Use it to answer the questions that follow.

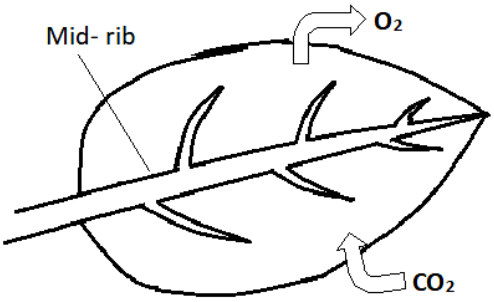

Figure 1

a. What process taking place in the leaf leads to this gaseous exchange shown in the **Figure 1** above?

_____

(1 mark)

b. Mention any **two** adaptations of a leaf to photosynthesis.

i. _____

_____

ii. _____

_____

(2 marks)

c. What is the main function of the mid-rib?

_____

_____

(1 mark)

d. Mention the function of the following cell organelles.

(i) Endoplasmic reticulum

_____

(1 mark)

(ii) Ribosomes

_____

(1 mark)

(iii) Mitochondrion

_____

(1 mark)

e. How do plants synthesize proteins?

_____

_____

_____

(2 marks)

2. **Figure 2** shows a bar graph of the human blood groups. Use it to answer the questions that follow.

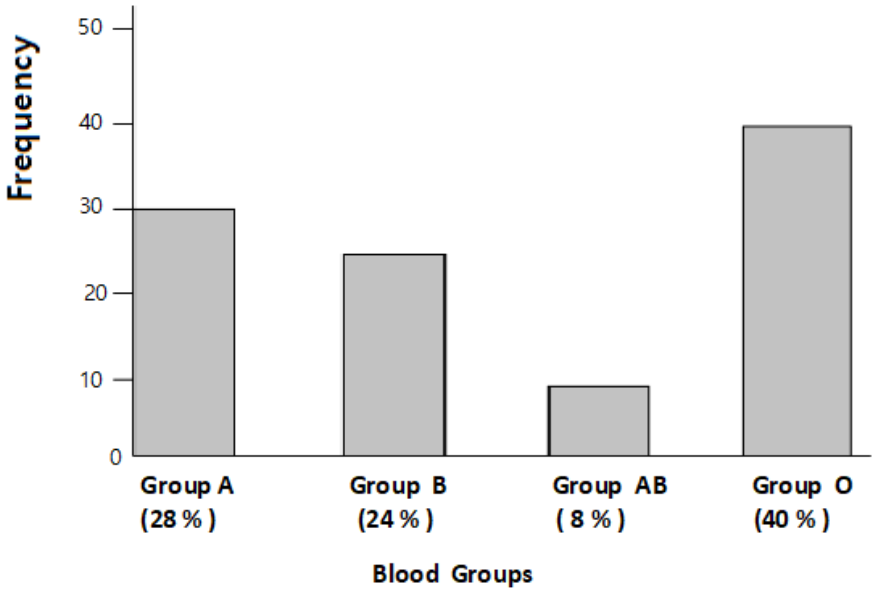

Figure 2

a. (i). What type of variation is shown by blood groups in the graph above?

_____

(1 mark)

(ii). Give a reason for your answer to **2 (a)(i)** above.

_____

_____

(2 marks)

b. In a population of 200 people, what would be the number of people with blood group **B**?

_____

_____

_____

(3 marks)

3. a. Define a "conditioned reflex action"

_____

_____

(1 mark)

b. What is the difference between a "conditioned reflex" and a "simple reflex action"?

_____

_____

(2 marks)

c. State any **two** abnormal conditions of the nervous system.

_____

_____

(2 marks)

4. **Figure 3** is a diagram showing a protozoa. Use it to answer the questions that follow.

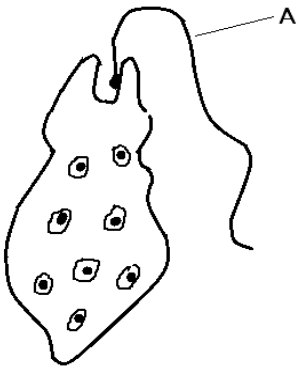

Figure 3

a. (i). Name the part labelled **A.**

_____

(1 mark)

(ii). What is the function of the part marked **A**?

_____

_____

(2 marks)

b. Explain any **four** roles played by micro-organisms in biotechnology.

_____

_____

_____

_____

_____

_____

_____

_____

(4 marks)

5. a. Define the term hypothesis.

_____

(1 mark)

b. Why is it important to report an investigation?

_____

_____

(2 marks)

c. Explain the significance of a control variable in a scientific investigation.

_____

_____

(2 marks)

6. **Figure 4 is** a diagram showing a vessel of the circulatory system. Use it to answer the questions that follow.

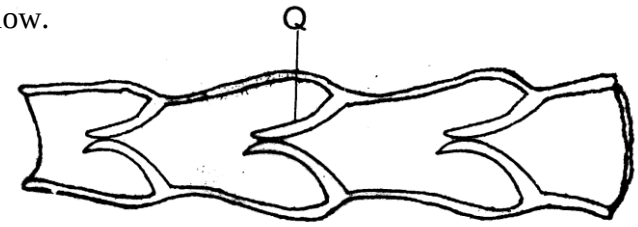

Figure 4

a. Identify the structure marked **Q**.

_____

(1 mark)

b. Name the other **two** vessels in the human body which have the structure labelled **Q.**

(i) _____

(ii) _____

(2 marks)

c. On the diagram above, put an arrow that indicates the direction of flow of blood.

d. Describe what would likely to happen if structure **Q** were damaged.

_____

_____

(2 marks)

e. State any **two** factors that would enable blood to move along the vessel.

_____

_____

(2 marks)

7. **Figure 5** is a diagram of an apparatus used to demonstrate the action of breathing mechanism in humans. Use it to answer the questions that follow.

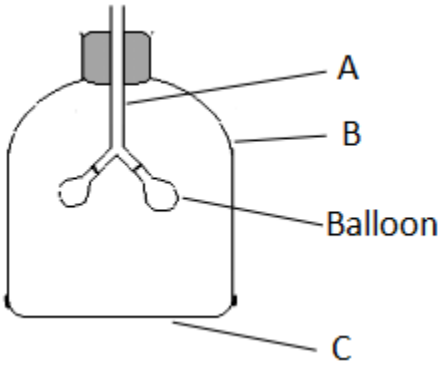

Figure 5

a. Name the parts of the real respiratory system that correspond to the parts labelled **A** and **C** in the figure above.

A: _____

C: _____

(2 marks)

b. Explain how the apparatus in **Figure 5** above demonstrates the breathing mechanism in human beings.

_____

_____

_____

_____

_____

_____

_____

(5 marks)

c. In what way does the part labelled **B** fail to demonstrate the action of breathing that the real lung demonstrates?

_____

_____

(2 marks)

8. **Figure 6** below is a diagram showing the carbon cycle. Use it to answer the questions that follow.

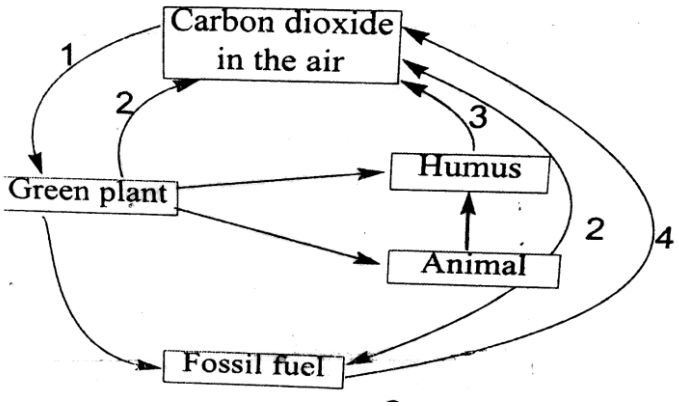

Figure 6

a. Name the processes marked **1**, **2** and **4** above.

1: _____

2: _____

4: _____

(3 marks)

b. Mention **two** groups of organisms that play a major role in process **3** above.

(i) _____

(ii) _____

(2 marks)

c. Name any **one** common fossil fuel.

_____

(1 mark)

d. Suggest any **one** possible way how deforestation could affect the carbon cycle.

_____

_____

(2 marks)

9. a. Define the following:

(i) Genetic engineering

_____

_____

(ii) Gene therapy

_____

_____

(2 marks)

b. Give any **two** ethical issues in biotechnology.

_____

_____

(2 marks)

c. Explain the importance of genetics during blood transfusion.

_____

_____

(2 marks)

10. a. What is the importance of peristalsis in the human digestive system?

_____

_____

(2 marks)

b. State any **two** roles played by the liver in relation to digestion.

_____

_____

(2 marks)

c. Explain any **two** functions of hydrochloric acid produced in the human stomach.

_____

_____

(2 marks)

SECTION B (30 marks)

Answer **all** the **three** questions in this section. All your answers should be in an essay form.

11. Discuss **five** ways in which a bird is adapted to overcoming gravity and the effects of drag in flight. Your answer should be in an essay form.

(10 marks)

12. Describe an experiment that can be conducted in order to investigate the effect of exercise on breathing rate. Your answer should be in an essay form.

(10 marks)

13. Describe any **five** ways in which breast feeding is important. Your answer should be in an essay form.

(10 marks)

END OF QUESTION PAPER

NB: This paper contains 11 printed pages